

## SIMATS ENGINEERING ASSESSMENT TOOL 1

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA1714

### **COURSE OUTCOME COVERED**

**CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

**Time Duration:** 1 Hour

**QUESTIONS:** 4

**Total Marks:** 40

Annexure A

### **CASE STUDY**

**Code of Conduct:**

I S. Mithesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

*S. Mithesh*

(90)

*AP*

CO3 - A41

## Artificial Intelligence

Q1. Cybase Intrusion Detection System

a) Propositional Logic Representation

F: Failed login Attempts  
U: unusual IP Address

O: odd login time

H: High Data transfer

B: Possible Brute Force Attacks

A: Possible Account Compromise

D: Possible Data Exfiltration

Rules:

$R_1: (F \wedge U) \rightarrow B$

$R_2: (F \wedge O) \rightarrow A$

$R_3: (U \wedge H) \rightarrow D$

$F_1: F = \text{True}$

$F_2: U = \text{True}$

$F_3: H = \text{True}$

b) Forward Chaining

Initial known facts  $\{F, U, H\}$

Iteration 1:

\* Rule  $R_1$  check:

$$R_1: (F \wedge U) \rightarrow B$$

• facts derived: B (Brute Force Attack)

• update Facts:  $\{F, U, H, B\}$ ;

\*  $R_3$  check: Antecedents U and H are both True

$$\text{Rule Fired: } R_3: (U \wedge H) \rightarrow D$$

Fact Derived: D (Data Exfiltration)

Update Facts:  $\{F, U, H, B, D\}$

Iteration 2:

Rule  $R_2$  check: Antecedent D is not in the known facts (False / unknown).  $R_2$  cannot fire

• Final Alerts:

Possible Brute Force Attack (B)

Possible Data Exfiltration (D)

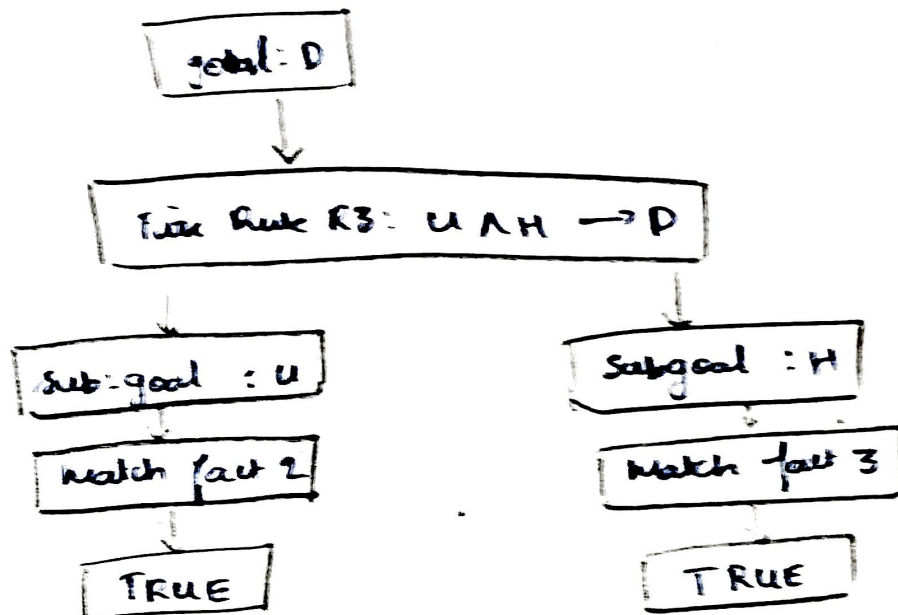
c) Backward chaining for goal: D

sub-goal: find rule & prove U & H

U: U matches fact 2 (U = True). (Proven)

H: H matches fact 3 (H = True). (Proven)

Conclusion: Both antecedents & verified via facts confirming D is True.



Q2 Airport Priority landing system

a) unification & mgu

Given Rule 1:  $\forall x : \text{Aircraft}(x) \wedge \text{Emergency Declared}(x)$

- Aircraft (Flight 2)
- Emergency Declared (Flight 2)

↓  
Priority Clearance (x)

unification:

1. match Antecedent Aircraft (x) with fact Aircraft (Flight 2)  
 $\theta_1 = \{x / \text{Flight 2}\}$

2. sub  $\theta_1$  int Emergency Declared (x):

Emergency Declared (Flight 2)

MCU (control unit)  $\theta = \{x / \text{Flight } x\}$

Desired Fact: Priority Clearance (Flight  $z$ )

b) Resolution for Goal:

1. From Rule 1 & unification

$c_1$ : Priority Clearance (Flight  $z$ )

2. From Rule 2 ( $\forall x$ : Priority Clearance ( $x$ ))

$\rightarrow$  Runway Reserved ( $x$ )  $\wedge$   $\neg$  Hold Pattern ( $x$ ):

$\neg$  Priority Clearance ( $x$ )  $\vee$  Runway Reserved ( $x$ )  
 $\wedge$   $\neg$  Hold Pattern ( $x$ )

$c_2$ :  $\neg$  Priority Clearance ( $x$ )  $\vee$  Runway Reserved ( $x$ )

$c_3$ :  $\neg$  Priority Clearance ( $x$ )  $\vee$   $\neg$  Hold Pattern ( $x$ )

3. From Rule 5 ( $\forall x \forall y$ : Aircraft ( $x$ )  $\wedge$  Queued After ( $x, y$ )  $\wedge$  Runway Reserved ( $y$ )  $\rightarrow$  Delay ( $x$ ):

$c_4$ :  $\neg$  Aircraft ( $x$ )  $\vee$   $\neg$  Queued After ( $x, y$ )  $\vee$   $\neg$  Runway Reserved ( $y$ )  
 $\vee$  Delay ( $x$ )

4.  $c_5$ : Aircraft (Flight  $y$ )

5.  $c_6$ : Queued After (Flight  $y$ , Flight  $z$ )

5. Negated goal:

$c_7$ :  $\neg$  Delay (Flight  $y$ )



a) CNF conversion

$C_1$ :  $\neg \text{MachineOverheating} \vee \text{MaintenanceNeeded}$

$C_2$ :  $\neg \text{MaintenanceNeeded} \vee \neg \text{HighPriorityLine} \vee \text{ScheduleImmediateRepair}$ .

$C_3$ :  $\text{ScheduleImmediateRepair} \vee \neg \text{HighPriorityLine} \vee \text{ProductionRisk}$

$C_4$ :  $\text{MachineOverheating}$ .

$C_5$ :  $\text{HighPriorityLine}$ .

b) Resolution for Immediate Repair.

• Negated goal ( $C_6$ ):  $\neg \text{ScheduleImmediateRepair}$

steps:

1.  $C_1 + C_6 \rightarrow \text{MaintenanceNeeded}$ .

2.  $C_2 + \text{MaintenanceNeeded} + C_5 \rightarrow \text{ScheduleImmediateRepair}$

3.  $\text{ScheduleImmediateRepair} + C_6 \rightarrow \text{contradiction}$

c) Removal

• without "MachineOverheating", "MaintenanceNeeded" & "ImmediateRepair".

• Does "ProductionRisk"